CEC 315 Exam Study Guide

Lectures 9–15: CT Fourier Series, Convergence, Properties, DT Fourier Series,
Frequency Response, Filtering, Fourier Transforms, Convolution, Magnitude, Phase, Bode Plots, and Systems

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This guide covers the main formulas, definitions, and conceptual explanations from Lectures 9–15—focus is on the "what" and "why" for exam preparation.

1 Lecture 9: LTI Eigenfunctions and the CT Fourier Series

1.1 The Central Idea

We previously decomposed signals as sums of shifted impulses, leading to convolution. Now we use a different decomposition: sums of **complex exponentials**. This leads to the **Fourier series**—a tool that breaks a periodic signal into its frequency components, like a prism splitting white light into colors.

Why complex exponentials? Because they are **eigenfunctions** of LTI systems.

1.2 Complex Exponentials as Eigenfunctions

An **eigenfunction** of a system is a special input that comes out as itself multiplied by a constant (the **eigenvalue**).

CT Eigenfunction Property

For a CT LTI system with impulse response $h(t)$, if the input is $x(t) = e^{st}$, the output is:

$$y(t) = H(s) \cdot e^{st}, \qquad H(s) = \int_{-\infty}^{+\infty} h(\tau)\, e^{-s\tau}\, d\tau$$

The system only changes the **amplitude and phase** of the exponential—never its frequency or shape.

DT Eigenfunction Property

For a DT LTI system with impulse response $h[n]$, if the input is $x[n] = z^n$:

$$y[n] = H(z) \cdot z^n, \qquad H(z) = \sum_{k=-\infty}^{+\infty} h[k]\, z^{-k}$$

The strategy: (1) Decompose the input into complex exponentials. (2) Multiply each component by the system's eigenvalue at that frequency. (3) Add them back up to get the output.

1.3 Periodic Signals and Harmonics

A CT signal is periodic with period T if $x(t) = x(t+T)$ for all t. The fundamental frequency is $\omega_0 = 2\pi/T$ (rad/s), or equivalently $f_0 = 1/T$ (Hz).

The **harmonically related complex exponentials** that share period T are:

$$\phi_k(t) = e^{jk\omega_0 t}, \qquad k = 0, \pm 1, \pm 2, \ldots$$

- $k = 0$: DC component (constant).

- $k = \pm 1$: fundamental (1st harmonic) at frequency ω_0.
- $k = \pm N$: Nth harmonic at frequency $N\omega_0$.

1.4 The Fourier Series: Synthesis and Analysis

CT Fourier Series Pair

Synthesis (build the signal from harmonics):

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k \, e^{jk\omega_0 t}$$

Analysis (find coefficients from the signal):

$$a_k = \frac{1}{T} \int_T x(t) \, e^{-jk\omega_0 t} \, dt$$

where $\int_T$ means integrate over any interval of length T.

The analysis equation works because of the **orthogonality** of complex exponentials:

$$\int_T e^{j(k-n)\omega_0 t} \, dt = \begin{cases} T, & k = n \\ 0, & k \neq n \end{cases}$$

Multiplying $x(t)$ by $e^{-jn\omega_0 t}$ and integrating "picks out" only the nth coefficient.

The DC Coefficient

$$a_0 = \frac{1}{T} \int_T x(t) \, dt \quad \text{(the average value of } x(t) \text{ over one period)}$$

1.5 Properties of Fourier Coefficients for Real Signals

If $x(t)$ is real, the coefficients satisfy **conjugate symmetry**:

$$a_{-k} = a_k^*$$

This means $|a_{-k}| = |a_k|$ (magnitude is even) and $\angle a_{-k} = -\angle a_k$ (phase is odd).

1.6 Trigonometric Form

For real signals, the complex exponential form can be rewritten as:

$$x(t) = a_0 + 2 \sum_{k=1}^{\infty} |a_k| \cos(k\omega_0 t + \angle a_k)$$

This is equivalent to the complex form but expressed entirely in terms of real cosines.

1.7 Key Euler Identities (Memorize These)

$$e^{j\theta} = \cos\theta + j\sin\theta, \qquad \cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}, \qquad \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}, \qquad \frac{1}{2j} = -\frac{j}{2}$$

Key Insight

Trick for simple signals: When $x(t)$ is given as a sum of sines and cosines, expand each term using Euler's relation and read off the coefficients by inspection—no integration needed.

Common Mistake

- a_0 must be computed separately (the general a_k formula is $0/0$ at $k = 0$).
- Don't confuse ω_0 (rad/s) with f_0 (Hz): $\omega_0 = 2\pi f_0$.
- Don't forget the $1/T$ factor in the analysis integral.
- The sum is infinite ($k = -\infty$ to $+\infty$); only simple signals have finitely many nonzero a_k.

2 Lecture 10: Convergence, Properties of CT Fourier Series, and DT Fourier Series

2.1 Convergence: When Does the Fourier Series Work?

Not every function has a valid Fourier series representation. The **Dirichlet conditions** give the mathematical guarantee.

Dirichlet Conditions for Convergence

A periodic signal $x(t)$ with period T has a convergent Fourier series if it satisfies **all three**:

1. **Absolute integrability:** $\int_T |x(t)|\, dt < \infty$ (every a_k is finite).
2. **Bounded variation:** Finitely many maxima and minima per period (no infinitely fast oscillations).
3. **Finite discontinuities:** Finitely many discontinuities per period, each with a finite jump.

Key Insight

Practical takeaway: Every periodic signal you encounter in this course—square waves, sawtooth waves, pulse trains, sinusoids—satisfies the Dirichlet conditions. Signals that violate them are mathematically pathological and essentially never appear in real systems.

2.2 The Gibbs Phenomenon

When a periodic signal has jump discontinuities:

- The Fourier series converges to the **midpoint of the jump** at each discontinuity. (Example: a square wave jumping from 0 to 1 converges to 0.5 at the jump.)
- A **truncated** Fourier series (finite number of harmonics) shows ripples near discontinuities.
- As more harmonics are added, the ripples get **narrower** and compress toward the discontinuity.
- However, the **peak overshoot remains at approximately 9%** of the jump height, no matter how many terms are included.
- In the limit of infinitely many terms the ripples become infinitely narrow and the error energy goes to zero.

Common Mistake

The $\approx 9\%$ overshoot is **not a bug**—it is a fundamental property of how sinusoids approximate sharp jumps. In applications like audio processing, this overshoot can matter and must be accounted for.

2.3 Properties of the CT Fourier Series

These properties let you find Fourier coefficients of complex signals from simpler ones **without re-integrating from scratch**. Notation: $x(t) \overset{\text{FS}}{\longleftrightarrow} a_k$.

Property	**Time Domain**	**FS Coefficients**
Linearity	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time Shifting	$x(t - t_0)$	$a_k\, e^{-jk\omega_0 t_0}$
Time Reversal	$x(-t)$	a_{-k}
Differentiation	$\dfrac{dx(t)}{dt}$	$jk\omega_0\, a_k$
Parseval's Relation	$\dfrac{1}{T}\displaystyle\int_T \lvert x(t)\rvert^2\, dt$	$= \displaystyle\sum_{k=-\infty}^{+\infty} \lvert a_k\rvert^2$

What Each Property Means

- **Linearity:** The coefficients of a linear combination are the same linear combination of the individual coefficients.
- **Time Shifting:** Delaying a signal multiplies each coefficient by a phase factor $e^{-jk\omega_0 t_0}$. The **magnitude spectrum $|a_k|$ is unchanged**—only the phase changes.
- **Time Reversal:** Reversing a signal in time reverses the order of the coefficients. For real even signals, a_k are purely real; for real odd signals, a_k are purely imaginary.
- **Differentiation:** Multiplies the kth coefficient by $jk\omega_0$, which amplifies higher harmonics. **Practical use:** if the derivative's coefficients are easier to find, compute those first and divide by $jk\omega_0$ (for $k \neq 0$). Always compute a_0 separately.
- **Parseval's Relation:** Average power of the signal equals the sum of powers in all harmonics. $|a_k|^2$ is the average power in the kth harmonic—no energy is lost or created.

Key Insight

Differentiation shortcut for triangular waves: The derivative of a triangular wave is a square wave. Find the square wave's coefficients d_k, then $a_k = d_k/(jk\omega_0)$ for $k \neq 0$. Result: triangular-wave coefficients decay as $1/k^2$ (much faster than the square wave's $1/k$) because the triangular wave is smoother (continuous, no jumps).

Key Insight

Parseval sanity check: Compute average power of $x(t)$ directly (left side) and compare with $\sum |a_k|^2$ (right side). If they don't match, recheck your coefficients!

2.4 The Discrete-Time Fourier Series

Periodicity in Discrete Time

A DT signal $x[n]$ is periodic with period N if $x[n] = x[n+N]$ for all n. The fundamental frequency is $\omega_0 = 2\pi/N$.

Only N Distinct Harmonics

In DT, complex exponentials that differ in frequency by a multiple of 2π are **identical**:

$$e^{j(k+N)(2\pi/N)n} = e^{jk(2\pi/N)n} \cdot \underbrace{e^{j2\pi n}}_{=1} = e^{jk(2\pi/N)n}$$

So there are only N distinct harmonics: $\phi_0[n], \phi_1[n], \ldots, \phi_{N-1}[n]$.

DT Fourier Series Pair

Synthesis:

$$x[n] = \sum_{k=\langle N\rangle} a_k \, e^{jk(2\pi/N)n}$$

Analysis:

$$a_k = \frac{1}{N} \sum_{n=\langle N\rangle} x[n] \, e^{-jk(2\pi/N)n}$$

where $\sum_{k=\langle N\rangle}$ means sum over any N consecutive values (e.g., $k = 0, 1, \ldots, N-1$).

Key Insight

Critical CT vs. DT difference:

- **CT:** Infinitely many distinct harmonics $\to$ infinite series. Convergence issues exist (Dirichlet, Gibbs).
- **DT:** Only N distinct harmonics $\to$ finite series with exactly N terms. **No convergence problems, no Gibbs phenomenon.**

DT Fourier coefficients are periodic: $a_{k+N} = a_k$ always. So a_{-1} and a_{N-1} are the **same** coefficient.

The analysis equation works because of **DT orthogonality**:

$$\sum_{n=\langle N\rangle} e^{jk(2\pi/N)n} = \begin{cases} N, & k = 0, \pm N, \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$$

2.5 Properties of the DT Fourier Series

Property	**Time Domain**	**FS Coefficients**
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time Shifting	$x[n - n_0]$	$a_k\, e^{-jk(2\pi/N)n_0}$
First Difference	$x[n] - x[n-1]$	$(1 - e^{-jk(2\pi/N)})\, a_k$
Parseval (DT)	$\frac{1}{N} \sum_{n=\langle N \rangle} \lvert x[n] \rvert^2$	$= \sum_{k=\langle N \rangle} \lvert a_k \rvert^2$
Multiplication	$x[n] \cdot y[n]$	$\sum_{l=\langle N \rangle} a_l\, b_{k-l}$

- **First Difference** is the DT analog of differentiation.
- **Multiplication** in time corresponds to **periodic convolution** of the Fourier coefficients.
- **Parseval (DT):** The sum on the right is **finite** (N terms), unlike the CT case.

2.6 CT vs. DT Fourier Series Comparison

	Continuous Time	**Discrete Time**
Period	T (any positive real)	N (positive integer)
Fund. frequency	$\omega_0 = 2\pi/T$	$\omega_0 = 2\pi/N$
Synthesis	$\sum_{k=-\infty}^{+\infty} a_k\, e^{jk\omega_0 t}$	$\sum_{k=\langle N \rangle} a_k\, e^{jk(2\pi/N)n}$
Analysis	$\frac{1}{T} \int_T x(t)\, e^{-jk\omega_0 t}\, dt$	$\frac{1}{N} \sum_{n=\langle N \rangle} x[n]\, e^{-jk(2\pi/N)n}$
Number of harmonics	Infinite	N (finite)
Convergence issues?	Yes (Dirichlet, Gibbs)	No
Coefficients periodic?	No	Yes, with period N

Common Mistake

- Do **not** write $\sum_{k=-\infty}^{+\infty}$ for the DT synthesis—it has only N terms.
- DT coefficients satisfy $a_k = a_{k+N}$ always, so $a_{-1} = a_{N-1}$.
- CT analysis uses an **integral** divided by T; DT analysis uses a **sum** divided by N. Do not swap them.
- When using the differentiation (CT) or first-difference (DT) property, always compute a_0 **separately** from the average value.

3 Lecture 11: Frequency Response and Filtering

3.1 The Central Idea

Lectures 9–10 showed how to decompose periodic signals into sums of complex exponentials (Fourier series). Lecture 6 showed that the output of an LTI system is found by convolution. **Now we combine these:** when the input to an LTI system is periodic, each frequency component is modified *independently* by the system. This leads to the **frequency response**—the foundation of filtering.

The logic rests on three building blocks:

1. **Eigenfunction property** (Lecture 9): $e^{jk\omega_0 t}$ through an LTI system yields $H(jk\omega_0)\, e^{jk\omega_0 t}$.
2. **Fourier series** (Lectures 9–10): $x(t) = \sum_k a_k\, e^{jk\omega_0 t}$.
3. **Superposition** (Lecture 6): The response to a sum is the sum of individual responses.

3.2 Frequency Response Definitions

CT Frequency Response

For a CT LTI system with impulse response $h(t)$:

$$H(j\omega) = \int_{-\infty}^{+\infty} h(\tau)\, e^{-j\omega\tau}\, d\tau$$

$|H(j\omega)|$ = how much the system amplifies or attenuates at frequency ω.
$\angle H(j\omega)$ = how much the system shifts the phase at frequency ω.

DT Frequency Response

For a DT LTI system with impulse response $h[n]$:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n]\, e^{-j\omega n}$$

Key property: $H(e^{j\omega})$ is **periodic in** ω **with period** 2π, because $e^{j(\omega+2\pi)n} = e^{j\omega n}$. Low frequencies are near $\omega = 0$; high frequencies are near $\omega = \pm\pi$.

3.3 Output of an LTI System for Periodic Input

The Frequency-by-Frequency Multiplication Rule

CT: If $x(t) = \sum_k a_k\, e^{jk\omega_0 t}$, then the output is:

$$y(t) = \sum_{k=-\infty}^{+\infty} a_k\, H(jk\omega_0)\, e^{jk\omega_0 t}, \qquad \boxed{b_k = a_k \cdot H(jk\omega_0)}$$

DT: If $x[n] = \sum_{k=\langle N\rangle} a_k\, e^{jk(2\pi/N)n}$, then:

$$y[n] = \sum_{k=\langle N\rangle} a_k\, H\big(e^{j2\pi k/N}\big)\, e^{jk(2\pi/N)n}, \qquad \boxed{b_k = a_k \cdot H\big(e^{j2\pi k/N}\big)}$$

Each input coefficient a_k is multiplied by the frequency response evaluated at the corresponding harmonic frequency.

Key Insight

The big picture: An LTI system acts as a **frequency-by-frequency multiplier**. It independently scales and phase-shifts each harmonic. The frequency response $H(j\omega)$ (or $H(e^{j\omega})$) is the "recipe" for how each frequency is treated.

3.4 Introduction to Filtering

Filtering is the process of modifying the frequency content of a signal. Since $b_k = a_k \cdot H(jk\omega_0)$, the frequency response determines which frequencies are passed and which are attenuated.

Two Categories of Filters

- **Frequency-shaping filters:** Change the relative amplitudes of different components. Examples: audio equalizers (boost bass or treble), differentiators ($H(j\omega) = j\omega$, amplifies highs).
- **Frequency-selective filters:** Pass certain frequency bands and reject others:
  - **Lowpass (LP):** Passes $|\omega| \le \omega_c$, rejects above.
  - **Highpass (HP):** Passes $|\omega| \ge \omega_c$, rejects below.
  - **Bandpass (BP):** Passes $\omega_{c1} \le |\omega| \le \omega_{c2}$, rejects outside.

  Terminology:
  - **Passband:** frequencies passed approximately without change.
  - **Stopband:** frequencies rejected (attenuated to ≈ 0).
  - **Cutoff frequency (ω_c):** boundary between passband and stopband.

Common Mistake

Ideal frequency-selective filters (with perfectly sharp transitions) **cannot be built in practice**. Real filters always have a gradual transition (roll-off) between passband and stopband.

3.5 The RC Lowpass Filter

The first-order RC circuit (input: source voltage $v_s(t)$; output: capacitor voltage $v_C(t)$) is described by $RC\,\dfrac{dv_C}{dt} + v_C = v_s$.

RC Lowpass Frequency Response

$$H(j\omega) = \frac{1}{1 + j\omega RC}$$

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}, \qquad \angle H(j\omega) = -\arctan(\omega RC)$$

Key observations:

- At $\omega = 0$ (DC): $|H| = 1$. Low frequencies pass through unattenuated.
- As $|\omega|$ increases: $|H|$ decreases toward 0. Higher frequencies are progressively attenuated.
- **Cutoff frequency:** $\omega_c = 1/(RC)$, where $|H(\omega_c)| = 1/\sqrt{2} \approx 0.707$ (power halved, -3 dB point).

3.6 The RC Highpass Filter

Using the **same** RC circuit but taking the **resistor voltage** $v_R(t)$ as output:

$$G(j\omega) = \frac{j\omega RC}{1 + j\omega RC}$$

At $\omega = 0$: $|G| = 0$ (DC blocked). As $|\omega| \to \infty$: $|G| \to 1$ (high frequencies pass).

Key Insight

The same physical RC circuit produces both a lowpass and a highpass filter depending on where you measure the output. Power is conserved: $|H(j\omega)|^2 + |G(j\omega)|^2 = 1$ at every frequency.

3.7 Time–Frequency Trade-Off

If we want a sharper lowpass filter (smaller ω_c, reject more high frequencies), we need a larger RC. But a larger RC makes the step response slower. **This trade-off is fundamental:** you cannot have both sharp frequency selectivity and fast time-domain response.

3.8 Discrete-Time Filtering

First-Order Recursive (IIR) Filter

The difference equation $y[n] - a\,y[n-1] = x[n]$ with $|a| < 1$ (stability) has frequency response:

$$H(e^{j\omega}) = \frac{1}{1 - a\,e^{-j\omega}}$$

- $0 < a < 1$: **lowpass** (passes near $\omega = 0$, attenuates near $\omega = \pi$).
- $-1 < a < 0$: **highpass** (passes near $\omega = \pi$, attenuates near $\omega = 0$).
- $|a|$ closer to 1: narrower passband but approaches instability.

Moving-Average (FIR) Filters

Two-point average (lowpass):

$$y[n] = \tfrac{1}{2}\big(x[n] + x[n-1]\big), \quad H(e^{j\omega}) = e^{-j\omega/2}\cos(\omega/2)$$

Passes DC perfectly ($H(e^{j0}) = 1$), completely rejects $\omega = \pi$ ($H(e^{j\pi}) = 0$).

Two-point difference (highpass):

$$y[n] = \tfrac{1}{2}\big(x[n] - x[n-1]\big), \quad H(e^{j\omega}) = je^{-j\omega/2}\sin(\omega/2)$$

Rejects DC ($H(e^{j0}) = 0$), passes $\omega = \pi$ ($|H(e^{j\pi})| = 1$).

Key Insight

FIR vs. IIR:

- **FIR (nonrecursive):** Finite-length impulse response $\Rightarrow$ **always stable**.
- **IIR (recursive):** Infinite impulse response (decays geometrically) $\Rightarrow$ can achieve sharper selectivity with fewer parameters, but **can be unstable** if $|a| \geq 1$.

Common Mistake

- Evaluate $H(j\omega)$ at the **harmonic frequencies** $k\omega_0$ only—not at arbitrary ω. Only those frequencies appear in the input.
- Do not confuse the frequency response $H(j\omega)$ (frequency domain) with the impulse response $h(t)$ (time domain). They are related by an integral transform but are different functions.
- The DT frequency response $H(e^{j\omega})$ repeats every 2π. Low DT frequencies are near $\omega = 0$; high DT frequencies are near $\omega = \pm\pi$.
- A single RC circuit gives a gentle roll-off, not a sharp cutoff. Sharper filtering requires higher-order systems.
- The frequency response $H(j\omega)$ is only well-defined for **stable** systems ($h(t)$ absolutely integrable).

4 Lecture 12: From Fourier Series to Fourier Transforms

4.1 The Central Idea

The Fourier series only handles **periodic** signals. Most real-world signals (a spoken word, a radar pulse) are **aperiodic**. The Fourier transform extends frequency-domain analysis to *any* signal with finite energy.

The derivation starts by embedding an aperiodic signal $x(t)$ inside a periodic extension with period T. As $T \to \infty$, the harmonic spacing $\omega_0 = 2\pi/T$ shrinks to zero, the discrete Fourier series coefficients merge into a continuous function, and the synthesis sum becomes an integral.

4.2 CT Fourier Transform Pair

CT Fourier Transform

Analysis (forward):

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t)\, e^{-j\omega t}\, dt$$

Synthesis (inverse):

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

The notation $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$ means these two are a **transform pair**.

Convergence (Dirichlet conditions): The FT exists if $x(t)$ is absolutely integrable ($\int |x(t)|\, dt < \infty$) and has a finite number of discontinuities per finite interval. Finite-energy signals also have a well-defined FT.

4.3 Basic CT Transform Pairs

Signal	$x(t)$	$X(j\omega)$
One-sided exponential	$e^{-at}u(t),\ a > 0$	$\dfrac{1}{a + j\omega}$
Two-sided exponential	$e^{-a\lvert t\rvert},\ a > 0$	$\dfrac{2a}{a^2 + \omega^2}$
Unit impulse	$\delta(t)$	1
Rectangular pulse (width $2T_1$)	$\begin{cases} 1, & \lvert t\rvert \le T_1 \\ 0, & \lvert t\rvert > T_1 \end{cases}$	$\dfrac{2\sin(\omega T_1)}{\omega}$
Complex exponential	$e^{j\omega_0 t}$	$2\pi\,\delta(\omega - \omega_0)$
Cosine	$\cos(\omega_0 t)$	$\pi\big[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)\big]$
Sine	$\sin(\omega_0 t)$	$\dfrac{\pi}{j}\big[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)\big]$

Key Insight

Sanity checks you should always apply:

- $X(0) = \int x(t)\,dt$ (the DC value equals the total area of the signal).
- If $x(t)$ is real and even $\Rightarrow X(j\omega)$ is real and even.
- If $x(t)$ is real and odd $\Rightarrow X(j\omega)$ is purely imaginary and odd.

4.4 Periodic Signals via the FT

A periodic signal $x(t) = \sum_k a_k\, e^{jk\omega_0 t}$ has a Fourier transform consisting of impulses:

$$X(j\omega) = \sum_{k=-\infty}^{+\infty} 2\pi\, a_k\, \delta(\omega - k\omega_0)$$

This bridges the Fourier series and the Fourier transform: the FS coefficients a_k become the **weights** of the impulses in $X(j\omega)$.

4.5 Discrete-Time Fourier Transform (DTFT)

DTFT Pair

Analysis:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]\, e^{-j\omega n}$$

Synthesis:

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

Key Insight

Critical difference from the CT FT: The DTFT $X(e^{j\omega})$ is **periodic in ω with period** 2π. This is because $e^{j(\omega+2\pi)n} = e^{j\omega n}$ for integer n. We only need to plot one period, typically $-\pi \leq \omega \leq \pi$.

Basic DTFT pairs:

Signal	$x[n]$	$X(e^{j\omega})$
Unit impulse	$\delta[n]$	1
DT exponential	$a^n u[n],\ \lvert a\rvert < 1$	$\dfrac{1}{1 - ae^{-j\omega}}$
Rectangular window (length $M+1$)	$\begin{cases} 1, & 0 \leq n \leq M \\ 0, & \text{else} \end{cases}$	$e^{-j\omega M/2}\, \dfrac{\sin\left(\frac{(M+1)\omega}{2}\right)}{\sin(\omega/2)}$

Convergence: The DTFT converges when $\sum |x[n]| < \infty$ (absolutely summable) or when $\sum |x[n]|^2 < \infty$ (finite energy).

5 Lecture 13: Fourier Transform Properties and Convolution

5.1 Why Properties Matter

Computing FT integrals from scratch is tedious. The properties let you **build new transform pairs from known ones** without re-integrating. The most important property—the **convolution property**—turns convolution in time into multiplication in frequency.

5.2 CT Fourier Transform Properties

Throughout, $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$ and $y(t) \overset{\mathcal{F}}{\longleftrightarrow} Y(j\omega)$.

Property	**Time Domain**	**Frequency Domain**
Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$
Frequency Shifting	$e^{j\omega_0 t} x(t)$	$X\big(j(\omega - \omega_0)\big)$
Conjugation	$x^*(t)$	$X^*(-j\omega)$
Time Reversal	$x(-t)$	$X(-j\omega)$
Time Scaling	$x(at)$	$\frac{1}{\lvert a\rvert} X\left(\frac{j\omega}{a}\right)$
Time Differentiation	$\frac{dx}{dt}$	$j\omega\, X(j\omega)$
nth Derivative	$\frac{d^n x}{dt^n}$	$(j\omega)^n\, X(j\omega)$
Freq. Differentiation	$-jt\, x(t)$	$\frac{dX(j\omega)}{d\omega}$
Convolution	$x(t) * h(t)$	$X(j\omega) \cdot H(j\omega)$
Multiplication	$x(t) \cdot y(t)$	$\frac{1}{2\pi} X(j\omega) * Y(j\omega)$
Duality	$X(jt)$	$2\pi\, x(-\omega)$
Parseval	$\int \lvert x(t)\rvert^2\, dt$	$= \frac{1}{2\pi} \int \lvert X(j\omega)\rvert^2\, d\omega$

What Each Property Means (Plain English)

- **Linearity:** Superposition holds—the FT of a sum is the sum of the FTs.
- **Time Shifting:** Delaying a signal by t_0 only changes the *phase* (by $-\omega t_0$); the magnitude spectrum is unchanged.
- **Frequency Shifting:** Multiplying by a complex exponential in time shifts the spectrum in frequency. This is the basis of **modulation** in communications.
- **Conjugate Symmetry (real signals):** If $x(t)$ is real, then $X(-j\omega) = X^*(j\omega)$. Consequences: $|X|$ is even, $\angle X$ is odd. Negative frequencies carry no new information.
- **Differentiation:** Taking a derivative in time is the same as multiplying by $j\omega$ in frequency. Higher frequencies get amplified—this is why derivatives enhance edges.
- **Time Scaling:** Compressing a signal in time ($|a| > 1$) stretches its spectrum and lowers its peak, and vice versa. This is the **time–frequency trade-off**.
- **Duality:** If you know one FT pair, you get a second one for free by swapping the roles of time and frequency.
- **Parseval:** Total signal energy can be computed in either domain. $|X(j\omega)|^2$ is the **energy spectral density**.

5.3 The Convolution Property

The Most Important Property

$$y(t) = x(t) * h(t) \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad Y(j\omega) = X(j\omega) \cdot H(j\omega)$$

Convolution in time becomes **multiplication** in frequency. At each frequency ω, the system scales the input by $|H(j\omega)|$ and shifts its phase by $\angle H(j\omega)$.

This is the payoff of frequency-domain analysis: instead of evaluating a difficult convolution integral, you transform to frequency, multiply, and transform back.

DT version: $y[n] = x[n] * h[n] \overset{\mathcal{F}}{\longleftrightarrow} Y(e^{j\omega}) = X(e^{j\omega}) \cdot H(e^{j\omega})$.

5.4 The Multiplication Property

Dual of Convolution

$$r(t) = x(t) \cdot y(t) \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad R(j\omega) = \frac{1}{2\pi} X(j\omega) * Y(j\omega)$$

Multiplication in time becomes **convolution** in frequency (with a $1/(2\pi)$ factor).

This underlies amplitude modulation, sampling, and windowing.

Common Mistake

Do **not** confuse convolution ($*$) with multiplication ($\cdot$). They swap directions:

- Convolution in time $\leftrightarrow$ multiplication in frequency.
- Multiplication in time $\leftrightarrow$ convolution in frequency.

The multiplication property has a $1/(2\pi)$ factor; the convolution property does **not**.

5.5 DTFT Properties (Side-by-Side)

Property	**CT FT**	**DTFT**
Time shift	$e^{-j\omega t_0}X$	$e^{-j\omega n_0}X$
Freq. shift	$X(j(\omega-\omega_0))$	$X(e^{j(\omega-\omega_0)})$
Differencing	$j\omega X$	$(1-e^{-j\omega})X$
Convolution	$X \cdot H$	$X \cdot H$
Multiplication	$\frac{1}{2\pi}X * Y$	$\frac{1}{2\pi}\int_{2\pi} X \cdot Y \, d\theta$
Parseval	$\int \lvert x\rvert^2 dt = \frac{1}{2\pi}\int \lvert X\rvert^2 d\omega$	$\sum \lvert x[n]\rvert^2 = \frac{1}{2\pi}\int_{2\pi} \lvert X\rvert^2 d\omega$

5.6 Systems from Differential and Difference Equations

The differentiation property lets you solve for $H(j\omega)$ algebraically.

CT: Differential Equation $\rightarrow$ Frequency Response

Given $\displaystyle\sum_{k=0}^{N} a_k \frac{d^k y}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x}{dt^k}$, replace d^k/dt^k with $(j\omega)^k$:

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum_{k=0}^{M} b_k\,(j\omega)^k}{\sum_{k=0}^{N} a_k\,(j\omega)^k}$$

DT: Difference Equation $\rightarrow$ Frequency Response

Given $\displaystyle\sum_{k=0}^{N} a_k\, y[n-k] = \sum_{k=0}^{M} b_k\, x[n-k]$, replace each delay ($n \rightarrow n-k$) with $e^{-jk\omega}$:

$$H(e^{j\omega}) = \frac{\sum_{k=0}^{M} b_k\, e^{-jk\omega}}{\sum_{k=0}^{N} a_k\, e^{-jk\omega}}$$

5.7 The Four-Step Analysis Pipeline

1. **Transform:** Find $H(j\omega)$ from the system equation and $X(j\omega)$ from FT tables/properties.
2. **Multiply:** $Y(j\omega) = X(j\omega) \cdot H(j\omega)$.
3. **Partial Fractions:** Decompose $Y(j\omega)$ into simple terms.
4. **Invert:** Use known inverse FT pairs to find $y(t)$.

6 Lecture 14: Magnitude, Phase, Group Delay, and Filters

6.1 Magnitude–Phase Representation

Every Fourier transform can be written in polar form:

$$X(j\omega) = |X(j\omega)|\, e^{j\angle X(j\omega)}$$

- $|X(j\omega)|$: magnitude spectrum (how much energy at each frequency).
- $\angle X(j\omega)$: phase spectrum (timing/structure of the signal).

For **real signals**, conjugate symmetry gives:

- $|X(j\omega)|$ is even: $|X(-j\omega)| = |X(j\omega)|$.
- $\angle X(j\omega)$ is odd: $\angle X(-j\omega) = -\angle X(j\omega)$.

6.2 Energy Spectral Density and Parseval

$|X(j\omega)|^2$ is the **energy spectral density**. Parseval's relation says:

$$\underbrace{\int_{-\infty}^{+\infty} |x(t)|^2\, dt}_{\text{total energy (time)}} = \frac{1}{2\pi} \underbrace{\int_{-\infty}^{+\infty} |X(j\omega)|^2\, d\omega}_{\text{total energy (frequency)}}$$

6.3 The Decibel (dB) Scale

$$|H(j\omega)|_{\text{dB}} = 20 \log_{10} |H(j\omega)|$$

$\lvert H\rvert$	dB	Meaning
1	0 dB	No change
$1/\sqrt{2} \approx 0.707$	-3 dB	Half-power point (cutoff)
0.1	-20 dB	Attenuated to 10%
0.01	-40 dB	Attenuated to 1%
10	$+20$ dB	Amplified $10\times$

The -3 dB point is the standard definition of **cutoff frequency**: $|H|^2 = 1/2$ (power halved).

6.4 Why Phase Matters

Key Insight

Phase carries structure; magnitude carries energy. Two signals with the same magnitude spectrum but different phase spectra can look completely different in time. If you scramble the phase while keeping the magnitude, the result looks like noise—even though it has the same energy at every frequency.

6.5 Linear Phase and Time Delay

A system with phase $\angle H(j\omega) = -\omega t_d$ (a linear function of ω) imposes a **constant delay** t_d on all frequency components.

- **If $|H| = 1$ everywhere:** The output is a perfect, undistorted delayed copy: $y(t) = x(t - t_d)$.
- **If $|H| \neq 1$:** The output is filtered and delayed, but there is no *phase distortion* because all components are still delayed equally.
- **Nonlinear phase:** Different frequencies are delayed by different amounts, causing waveform distortion even if $|H|$ is flat.

DT: If $\angle H(e^{j\omega}) = -\omega n_d$ with n_d an integer, the output is $y[n] = x[n - n_d]$ (pure sample delay).

6.6 Group Delay

Group Delay Definition

$$\tau(\omega) = -\frac{d}{d\omega}\left[\angle H(j\omega)\right]$$

It measures the effective delay experienced by a narrow band of frequencies centered at ω.

- **Linear phase** $\Rightarrow \tau(\omega) = t_d$ (constant) $\Rightarrow$ no phase distortion.
- **Nonlinear phase** $\Rightarrow \tau(\omega)$ varies with $\omega \Rightarrow$ phase distortion.
- For many applications (audio, communications), approximately **constant group delay** over the passband is as important as a flat magnitude.

Common Mistake

Do **not** confuse group delay with phase delay. Phase delay is $-\angle H/\omega$ (delay of the carrier). Group delay is $-d(\angle H)/d\omega$ (delay of the envelope). They are equal only when phase is perfectly linear.

6.7 Ideal Frequency-Selective Filters

Ideal CT Lowpass Filter

$$H_{\mathrm{LP}}(j\omega) = \begin{cases} e^{-j\omega t_d}, & |\omega| \le \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

Magnitude is a perfect rectangle; phase is linear (constant delay t_d).
Impulse response:

$$h(t) = \frac{\sin\big[\omega_c(t - t_d)\big]}{\pi(t - t_d)} = \frac{\omega_c}{\pi}\,\mathrm{sinc}\left(\frac{\omega_c(t - t_d)}{\pi}\right)$$

This sinc extends to $t = -\infty$, so $h(t) \neq 0$ for $t < 0$: the system is **noncausal** and therefore **physically unrealizable**. Its step response exhibits $\approx 9\%$ overshoot (Gibbs phenomenon), regardless of ω_c.

Key Insight

No physically realizable filter can have a perfectly sharp frequency cutoff. A rectangle in one domain requires an infinite sinc in the other. All real filters must have a gradual transition.

DT Ideal Lowpass Filter

$$H_{\mathrm{LP}}(e^{j\omega}) = \begin{cases} e^{-j\omega n_d}, & |\omega| \le \omega_c \\ 0, & \omega_c < |\omega| \le \pi \end{cases} \qquad \text{(periodic with period } 2\pi\text{)}$$

Impulse response: $h[n] = \dfrac{\sin\big[\omega_c(n - n_d)\big]}{\pi(n - n_d)}$, which is also noncausal and unrealizable.

6.8 Nonideal (Real) Filter Specifications

Since ideal filters are unrealizable, real filters are described by practical specifications:

Parameter	Meaning
Passband edge ω_p	Highest frequency guaranteed to meet passband spec
Stopband edge ω_s	Lowest frequency guaranteed to meet stopband spec
Cutoff frequency ω_c	-3 dB point ($\lvert H\rvert^2 = 1/2$); falls inside the transition band
Transition bandwidth $\Delta\omega = \omega_s - \omega_p$	Width of the roll-off region
Passband ripple δ_p	Max deviation from unity in the passband (dB)
Stopband attenuation δ_s	Min attenuation in the stopband (dB)

The ideal filter has $\omega_p = \omega_s = \omega_c$ (zero-width transition), which is unrealizable.

6.9 The Fundamental Trade-Offs

Bandwidth–Rise-Time Trade-Off

$$\omega_c \cdot t_r \approx \text{constant} \quad (\text{typically } \approx \pi \text{ to } 2\pi)$$

Wider bandwidth $\Rightarrow$ faster step response. Narrower bandwidth $\Rightarrow$ slower response.

The three-way design trade-off:

1. **Transition width** $\Delta\omega$: how sharp is the cutoff?
2. **Time-domain ringing:** how much overshoot/oscillation?
3. **Filter order/complexity:** how many coefficients or components?

Improving any one typically worsens at least one of the others. Filter design is about finding the best compromise.

7 Lecture 15: First/Second-Order Systems and Bode Plots

7.1 Why This Lecture Matters

Most practical systems—circuits, mechanical structures, control loops—are described by first- or second-order equations. This lecture derives standard-form frequency responses, introduces Bode plots for quick visualization, and connects CT and DT system analysis.

7.2 First-Order CT Systems

Standard First-Order Lowpass

Differential equation: $\dfrac{dy}{dt} + \omega_c\, y(t) = \omega_c\, x(t)$. Frequency response:

$$\boxed{H(j\omega) = \frac{1}{1 + j\omega/\omega_c}}$$

Key characteristics:

- **Time constant:** $\tau = 1/\omega_c$. Impulse response: $h(t) = \omega_c\, e^{-\omega_c t}\, u(t)$.
- **DC gain:** $H(0) = 1$.
- **Bandwidth:** The -3 dB frequency is ω_c, where $|H| = 1/\sqrt{2}$.
- **Magnitude:** $|H(j\omega)| = \dfrac{1}{\sqrt{1 + (\omega/\omega_c)^2}}$
- **Phase:** $\angle H(j\omega) = -\arctan(\omega/\omega_c)$

The complementary **first-order highpass** is $H_{\text{HP}} = \dfrac{j\omega/\omega_c}{1 + j\omega/\omega_c} = 1 - H_{\text{LP}}$.

7.3 Bode Plots

A **Bode plot** consists of magnitude (dB) and phase (degrees) vs. frequency on a **logarithmic** scale. The log axis reveals asymptotic behavior as straight lines.

Key Insight

Quick-sketch rules for a first-order pole at ω_c:

1. **Magnitude:** 0 dB for $\omega < \omega_c$, then slope -20 dB/decade for $\omega > \omega_c$. True curve is 3 dB below the asymptote at ω_c.
2. **Phase:** $0°$ for $\omega < \omega_c/10$, straight line from $0°$ to $-90°$ spanning one decade on each side of ω_c, then $-90°$ for $\omega > 10\omega_c$. True value is $-45°$ at ω_c.

For a first-order **zero** at ω_z: same shapes but with **positive** slopes ($+20$ dB/decade, phase $0° \to +90°$).

7.4 Second-Order CT Systems

Standard Second-Order System

Differential equation: $\dfrac{d^2y}{dt^2} + 2\zeta\omega_n \dfrac{dy}{dt} + \omega_n^2\, y = \omega_n^2\, x$. Frequency response:

$$\boxed{H(j\omega) = \frac{1}{1 - (\omega/\omega_n)^2 + j\,2\zeta(\omega/\omega_n)}}$$

where ω_n = natural frequency (rad/s) and ζ = damping ratio (dimensionless).

Magnitude (let $u = \omega/\omega_n$):

$$|H| = \frac{1}{\sqrt{(1-u^2)^2 + (2\zeta u)^2}}$$

Key values:

- DC ($\omega = 0$): $|H| = 1$.
- At $\omega = \omega_n$: $|H| = 1/(2\zeta)$. Exceeds unity when $\zeta < 0.5$.
- **Resonance peak** occurs at $\omega_r = \omega_n\sqrt{1-2\zeta^2}$ (exists only when $\zeta < 1/\sqrt{2} \approx 0.707$), with peak magnitude $|H|_{\max} = \dfrac{1}{2\zeta\sqrt{1-\zeta^2}}$.

Phase:

$$\angle H = -\arctan\left(\frac{2\zeta(\omega/\omega_n)}{1-(\omega/\omega_n)^2}\right)$$

Phase is always $-90°$ at $\omega = \omega_n$, regardless of ζ. Goes from $0°$ (low freq.) to $-180°$ (high freq.).

Damping regimes:

- $\zeta < 1$: **Underdamped**—oscillatory step response.
- $\zeta = 1$: **Critically damped**—fastest response without overshoot.
- $\zeta > 1$: **Overdamped**—sluggish, no oscillation.
- $\zeta = 1/\sqrt{2}$: **Butterworth** (maximally flat magnitude).

Step Response Specifications (Underdamped, $\zeta < 1$)

- **Percent overshoot:** $\%\text{OS} = 100\exp\left(\dfrac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$
- **Settling time** (2% criterion): $t_s \approx 4/(\zeta\omega_n)$
- **Peak time:** $t_p = \pi/\omega_d$, where $\omega_d = \omega_n\sqrt{1-\zeta^2}$ is the damped natural frequency.

Bode Plot of Second-Order System

- High-frequency asymptote: -40 dB/decade (two poles $\Rightarrow$ twice the first-order slope).
- Near ω_n: deviation depends on ζ. For $\zeta = 0.707$, deviation is $+3$ dB; for $\zeta = 0.1$, peak is $+14$ dB above asymptote.
- Phase: $0° \to -180°$, with $-90°$ always at ω_n.

7.5 DT Systems

First-Order DT Lowpass

$$H(e^{j\omega}) = \frac{1-a}{1-ae^{-j\omega}}, \qquad 0 < a < 1$$

Single pole at $z = a$. Stable when $|a| < 1$. DC gain is 1. The closer a is to 1, the narrower the passband.

Second-Order DT System

Complex conjugate poles at $z = re^{\pm j\theta}$ (stable when $r < 1$):

- Pole **radius** r controls damping (smaller r = more damped).
- Pole **angle** θ controls the resonance frequency ($\omega_r \approx \theta$).

7.6 CT vs. DT System Summary

	First-Order	**Second-Order**
CT standard form	$H = \dfrac{1}{1 + j\omega/\omega_c}$	$H = \dfrac{1}{1 - (\omega/\omega_n)^2 + j2\zeta\omega/\omega_n}$
CT rolloff	-20 dB/dec	-40 dB/dec
CT key params	ω_c	ω_n, ζ
CT resonance?	No	Yes, when $\zeta < 1/\sqrt{2}$
DT standard form	$H = \dfrac{1-a}{1-ae^{-j\omega}}$	Poles at $z = re^{\pm j\theta}$
DT stability	$\lvert a\rvert < 1$	$r < 1$
DT resonance freq.	N/A	$\omega \approx \theta$

Common Mistake

- $\omega_n \neq \omega_c$ in a second-order system (unless $\zeta = 1/\sqrt{2}$).
- Slope rules: ± 20 dB/dec per pole or zero; ± 40 dB/dec for second-order.
- $\zeta < 1$ does **not** always give a resonance peak in $|H|$; the peak exists only for $\zeta < 1/\sqrt{2}$.
- In DT, larger r = **less** damping. In CT, larger σ = **more** damping. Directions are "opposite."
- Bode plots **must** use a log frequency axis.

8 Quick-Reference Formula Sheet

CT Fourier Series

Synthesis	$x(t) = \sum_{k=-\infty}^{+\infty} a_k \, e^{jk\omega_0 t}$
Analysis	$a_k = \frac{1}{T} \int_T x(t) \, e^{-jk\omega_0 t} \, dt$
DC coefficient	$a_0 = \frac{1}{T} \int_T x(t) \, dt$
Conjugate symmetry	$a_{-k} = a_k^* \quad$ (real $x(t)$)
Trigonometric form	$x(t) = a_0 + 2 \sum_{k=1}^{\infty} \lvert a_k \rvert \cos(k\omega_0 t + \angle a_k)$
Orthogonality	$\int_T e^{j(k-n)\omega_0 t} \, dt = \begin{cases} T, & k = n \\ 0, & k \neq n \end{cases}$

CT Fourier Transform Pairs

$x(t)$	$X(j\omega)$
$e^{-at}u(t),\ a > 0$	$\frac{1}{a + j\omega}$
$te^{-at}u(t),\ a > 0$	$\frac{1}{(a + j\omega)^2}$
$e^{-a\lvert t\rvert},\ a > 0$	$\frac{2a}{a^2 + \omega^2}$
$\delta(t)$	1
1	$2\pi\delta(\omega)$
$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos(\omega_0 t)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
Rect. pulse (width $2T_1$)	$\frac{2\sin(\omega T_1)}{\omega}$
$\frac{\sin(Wt)}{\pi t}$ (sinc)	Ideal rect. $\begin{cases} 1, & \lvert\omega\rvert \leq W \\ 0, & \lvert\omega\rvert > W \end{cases}$

DT Fourier Series

Synthesis	$x[n] = \sum_{k=\langle N\rangle} a_k\, e^{jk(2\pi/N)n}$
Analysis	$a_k = \frac{1}{N} \sum_{n=\langle N\rangle} x[n]\, e^{-jk(2\pi/N)n}$
DC coefficient	$a_0 = \frac{1}{N} \sum_{n=\langle N\rangle} x[n]$
Coefficient periodicity	$a_{k+N} = a_k$
DT orthogonality	$\sum_{n=\langle N\rangle} e^{jk(2\pi/N)n} = \begin{cases} N, & k = 0, \pm N, \ldots \\ 0, & \text{otherwise} \end{cases}$
Parseval (DT)	$\frac{1}{N} \sum_{n=\langle N\rangle} \lvert x[n]\rvert^2 = \sum_{k=\langle N\rangle} \lvert a_k\rvert^2$

CT FS Properties

Property	**Time**	**Coefficients**
Linearity	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time shift	$x(t - t_0)$	$a_k\, e^{-jk\omega_0 t_0}$
Time reversal	$x(-t)$	a_{-k}
Differentiation	dx/dt	$jk\omega_0\, a_k$
Parseval	$\frac{1}{T} \int_T \lvert x\rvert^2\, dt$	$= \sum \lvert a_k\rvert^2$

Frequency Response and Filtering

CT frequency response	$H(j\omega) = \int_{-\infty}^{+\infty} h(\tau)\, e^{-j\omega\tau}\, d\tau$
DT frequency response	$H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n]\, e^{-j\omega n}$ (periodic, period 2π)
CT output coefficients	$b_k = a_k \cdot H(jk\omega_0)$
DT output coefficients	$b_k = a_k \cdot H(e^{j2\pi k/N})$
RC lowpass	$H(j\omega) = \dfrac{1}{1 + j\omega RC}, \quad \omega_c = 1/(RC)$
RC highpass	$G(j\omega) = \dfrac{j\omega RC}{1 + j\omega RC}, \quad \lvert H\rvert^2 + \lvert G\rvert^2 = 1$
DT 1st-order recursive	$H(e^{j\omega}) = \dfrac{1}{1 - a\, e^{-j\omega}}, \quad \lvert a\rvert < 1$ for stability

DTFT Pairs

$x[n]$	$X(e^{j\omega})$
$\delta[n]$	1
$a^n u[n],\ \lvert a\rvert < 1$	$\dfrac{1}{1 - ae^{-j\omega}}$
Rect. window $(0 \le n \le M)$	$e^{-j\omega M/2}\, \dfrac{\sin \frac{(M+1)\omega}{2}}{\sin(\omega/2)}$

Key Relationships

- CT system from diff. eq.: replace $d^k/dt^k \to (j\omega)^k$, solve for $H(j\omega) = Y/X$.
- DT system from diff. eq.: replace delay $n \to n-k$ with $e^{-jk\omega}$, solve for $H(e^{j\omega}) = Y/X$.
- Output spectrum: $Y(j\omega) = X(j\omega) \cdot H(j\omega)$.
- Output FS coefficients: $b_k = a_k \cdot H(jk\omega_0)$ (CT) or $b_k = a_k \cdot H(e^{j2\pi k/N})$ (DT).
- Group delay: $\tau(\omega) = -d(\angle H)/d\omega$.
- -3 dB cutoff: $|H(\omega_c)|^2 = 1/2$.
- First-order rolloff: -20 dB/decade. Second-order: -40 dB/decade.
- 2nd-order resonance peak: exists when $\zeta < 1/\sqrt{2}$, at $\omega_r = \omega_n\sqrt{1 - 2\zeta^2}$.
- Percent overshoot: $\%\text{OS} = 100\exp(-\pi\zeta/\sqrt{1-\zeta^2})$. Settling time: $t_s \approx 4/(\zeta\omega_n)$.
- Eigenfunction property: $e^{st} \xrightarrow{\text{LTI}} H(s)\, e^{st}$.
- RC lowpass cutoff: $\omega_c = 1/(RC)$. Same circuit, resistor output $\to$ highpass.
- DT coefficient periodicity: $a_{k+N} = a_k$, so $a_{-1} = a_{N-1}$.

- Gibbs overshoot: $\approx 9\%$ of jump height (persists for any finite truncation).
- Triangular-wave coefficients decay as $1/k^2$ vs. square-wave $1/k$ (smoother signal $\rightarrow$ faster decay).

9 Common Mistakes to Avoid (All Lectures)

1. **Computing a_0 from the general a_k formula.** The general formula has k in the denominator and gives $0/0$ at $k = 0$. Always use $a_0 = \frac{1}{T}\int_T x(t)\,dt$ separately.

2. **Confusing ω_0 and f_0.** $\omega_0 = 2\pi/T$ (rad/s) vs. $f_0 = 1/T$ (Hz). Related by $\omega_0 = 2\pi f_0$.

3. **Confusing FS coefficients (a_k, discrete) with the FT ($X(j\omega)$, continuous).** They are related by $a_k = \frac{1}{T}X(jk\omega_0)$ for periodic signals, but they are different objects.

4. **Forgetting the $1/(2\pi)$ in the inverse FT.** The analysis equation has no $1/(2\pi)$; the synthesis equation does.

5. **Forgetting the DTFT is periodic in ω.** Unlike the CT FT, $X(e^{j\omega})$ repeats every 2π.

6. **Wrong sign in the exponent.** Forward transform: $e^{-j\omega t}$. Inverse: $e^{+j\omega t}$. Time delay $t_0 > 0$ gives $e^{-j\omega t_0}$.

7. **Swapping convolution and multiplication directions.** Conv. in time = mult. in freq. Mult. in time = conv. in freq. The multiplication property has a $1/(2\pi)$; the convolution property does not.

8. **Ignoring phase.** Phase carries structural/timing information. A flat magnitude with nonlinear phase still causes distortion.

9. **Confusing group delay with phase delay.** $\tau = -d(\angle H)/d\omega$ vs. $-\angle H/\omega$. Equal only for linear phase.

10. **Thinking ideal filters are realizable.** No physical system produces a perfectly sharp cutoff. The sinc impulse response is noncausal.

11. **Using the geometric series for $|a| \geq 1$.** $\sum_{n=0}^{\infty}(ae^{-j\omega})^n$ converges only when $|a| < 1$.

12. **Partial fractions in DT with the wrong variable.** Work with $e^{-j\omega}$ (or z^{-1}), not $e^{j\omega}$ or ω.

13. **Confusing ω_n with ω_c.** The natural frequency ω_n is **not** the -3 dB cutoff in a second-order system (unless $\zeta = 1/\sqrt{2}$).

14. **Assuming $\zeta < 1$ always gives a magnitude resonance peak.** The peak exists only when $\zeta < 1/\sqrt{2} \approx 0.707$.

15. **Plotting Bode plots on a linear frequency scale.** Bode plots require a logarithmic frequency axis for the straight-line approximation to work.

16. **Confusing DT pole radius with damping direction.** In DT, larger r = less damping. In CT, larger σ = more damping. Directions are opposite.

17. **Using infinite limits for the DT Fourier series synthesis.** The DT synthesis sum has only N terms. Do not write $\sum_{k=-\infty}^{+\infty}$ for the DT case.

18. **Forgetting that DT Fourier coefficients are periodic.** $a_k = a_{k+N}$ always. So a_{-1} and a_{N-1} are the **same** coefficient.

19. **Mixing up CT and DT analysis formulas.** CT uses an **integral** divided by T; DT uses a **sum** divided by N. Do not swap them.

20. **Ignoring the Gibbs phenomenon.** When approximating a discontinuous CT signal with finitely many harmonics, expect $\approx 9\%$ overshoot near discontinuities. The approximation will not be smooth there.

21. **Forgetting a_0 when using differentiation or first-difference properties.** These properties give a_k for $k \neq 0$ only. Always compute a_0 (the average value) separately.

22. **Evaluating $H(j\omega)$ at the wrong frequencies.** For a periodic input with fundamental frequency ω_0, evaluate H at the harmonic frequencies $k\omega_0$—not at arbitrary ω.

23. **Confusing $H(j\omega)$ with $h(t)$.** The frequency response $H(j\omega)$ lives in the frequency domain; the impulse response $h(t)$ lives in the time domain. Related by an integral transform, but different functions.

24. **Expecting ideal filter behavior from simple circuits.** A single RC circuit gives a gentle roll-off, not a sharp cutoff. Sharper filtering requires higher-order systems.

25. **Applying Fourier analysis to unstable systems.** $H(j\omega)$ is only well-defined for stable systems (absolutely integrable $h(t)$, or $|a| < 1$ in DT recursive filters).

Good luck on the exam!